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TECHNICAL DOCUMENTATION

Live Chat, Voice & Video Calling Platform

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Platform	Android / iOS (Flutter / React Native)

18+ Platform | Moderated | Verified Listeners Only



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1. Application Overview

This document describes the complete technical specification for a one-to-one live communication platform developed by **Satya Kabir E-Solutions Pvt. Ltd.** The application enables users (customers) to connect with verified female listeners via text chat, voice calls, and video calls on a paid-per-minute basis. The platform is an 18+ service with strong identity verification, moderation, and a coin-based economy.

Key Objectives

- Single mobile app supporting two distinct user roles: Customer and Listener
- Real-time one-to-one chat, voice call, and video call sessions
- Coin/credit-based monetisation with per-minute billing
- Listener identity verification and admin approval workflow
- Secure, scalable backend with real-time notification support
- Admin panel for full platform governance

High-Level Feature Summary

Feature	Description
Live Text Chat	One-to-one chat with emoji, stickers, voice messages
Voice Calling	Real-time audio call with per-minute billing
Video Calling	HD video call with auto-disconnect on low balance
Wallet System	Coin purchase, usage history, in-app top-up
Listener Earnings	Per-session earnings, withdrawal to bank/UPI
Admin Control	User management, listener verification, reports



2. System Architecture

The application follows a client-server architecture with dedicated real-time communication layers. The backend is decoupled into REST API services and real-time signalling services.

Architecture Layers

Layer	Technology	Responsibility
Mobile Client	Flutter / React Native	Android & iOS app for both roles
REST API	Node.js + Express	Authentication, profiles, wallet, history
Real-Time Layer	Socket.io / Agora / WebRTC	Chat signalling, call setup & teardown
Media Server	Agora RTC / Twilio	Audio & video stream relay
Database	MongoDB (primary)	Users, sessions, transactions, messages
Cache	Redis	Session tokens, online presence, rate limiting
File Storage	AWS S3 / Firebase Storage	Profile photos, ID documents
Notifications	Firebase Cloud Messaging	Push notifications for all events
Payments	Razorpay / Stripe	Coin purchase, listener payout
Admin Panel	React.js (Web)	Full platform management

Data Flow — Session Initiation

User selects listener → App checks wallet balance → REST API creates session record → Socket.io emits call invitation to listener → Listener accepts → Agora/WebRTC channel opened → Per-minute timer starts → Balance deducted every 60 seconds → Session ends on hangup or zero balance.





3. Authentication & User Roles

3.1 Customer Authentication

- Mobile OTP via SMS gateway (MSG91 / Twilio Verify) or Email OTP
- JWT access token (expiry: 1h) + Refresh token (expiry: 30d) stored securely
- Optional social login (Google)
- Profile: name, display photo, age (optional), preferred language

3.2 Listener Authentication

- Separate registration flow with enhanced verification
- Required fields: full name, mobile, date of birth (18+ check), government ID upload, live selfie
- Admin manually reviews and approves/rejects listener account
- On approval, listener receives email + push notification
- Listeners can set per-minute rates for chat, voice, and video independently

3.3 Role-Based Access Control

Role	Permissions
CUSTOMER	Browse listeners, initiate chat/call/video, manage wallet, view history
LISTENER	Receive requests, manage availability & rates, view earnings, withdraw
ADMIN	Full platform access: approve listeners, manage users, set commission, view reports
SUPER_ADMIN	System configuration, rate limits, feature flags, payout schedules



4. User (Customer) Module

4.1 Home Screen

The home screen presents a real-time grid/list of online listeners fetched via paginated REST API (cached in Redis for 15 seconds to reduce DB load).

- Online listener cards with photo, name, rating, and rate-per-minute
- Filter by: language, category (friendly, advice, casual), popularity, price range
- Search by listener name
- Real-time online status via Socket.io presence events
- Pull-to-refresh and infinite scroll pagination

4.2 Listener Profile View

- Profile photo gallery (up to 5 photos)
- Bio / Introduction text
- Languages spoken badges
- Chat rate, voice rate, video rate (per minute in coins)
- Average star rating and total sessions
- User reviews (most recent 10)
- Action buttons: Chat, Voice Call, Video Call

4.3 Chat Module

Feature	Detail
Text Message	Real-time text with delivery & read receipts
Emoji / Stickers	Built-in emoji picker + custom sticker packs
Voice Message	Record and send audio clips (max 60 seconds)
Escalate to Call	One-tap upgrade from chat to voice or video call
Billing	Per-minute coin deduction, shown in chat header
History	Full chat history with timestamps, persisted in DB

4.4 Voice Call Module

- One-to-one audio call using Agora Voice SDK or WebRTC
- In-call UI: listener avatar, call duration timer, live coin balance
- Mute / Speaker toggle
- Per-minute deduction with 10-second low-balance warning
- Auto-disconnect when balance reaches zero
- Post-call rating popup (1–5 stars + optional comment)
- Call duration and cost summary screen

4.5 Video Call Module



- One-to-one HD video call using Agora Video SDK
- Front camera default; camera switch supported
- Pip (picture-in-picture) self preview
- Video on/off and mic mute controls
- Per-minute billing identical to voice call logic
- Auto-disconnect and cost summary on session end





5. Listener / Girl Module

5.1 Dashboard

- Online / Offline / Busy status toggle (synced via Socket.io)
- Today's earnings summary widget
- Incoming request queue: chat requests, voice calls, video calls
- Accept / Reject buttons with 20-second timeout (auto-reject)
- Active session counter

5.2 Profile Management

- Update display name, bio, photos
- Set languages spoken (multi-select)
- Configure per-minute rates: chat (coins), voice (coins), video (coins)
- View verification badge status

5.3 Earnings & Withdrawal

Item	Detail
Earnings Wallet	Accumulated coins converted to INR at platform rate
Session History	Per-session breakdown: duration, coins earned, date
Reports	Daily / Weekly / Monthly earnings charts
Withdrawal	Request payout to linked bank account or UPI ID
Minimum Payout	Configurable by admin (default: ₹500)
Commission	Platform deducts commission % before payout (set by admin)

5.4 Safety Controls

- Block a user — prevents future contact
- Report abusive behaviour with category + description
- Optional: auto call-recording flag for safety review (admin access only)
- Emergency disconnect — end session immediately





6. Real-Time Communication

6.1 Socket.io Events

Event	Direction	Description
user:connect	Client → Server	User comes online
user:disconnect	Server → Clients	User goes offline; update presence
chat:send	Client → Server	Send chat message
chat:receive	Server → Client	Deliver message to recipient
call:request	Client → Server	Initiate voice/video call
call:incoming	Server → Listener	Notify listener of call
call:accept	Listener → Server	Listener accepts call
call:reject	Listener → Server	Listener rejects call
call:end	Client → Server	End call, trigger billing
balance:low	Server → User	Warn user of low balance
balance:zero	Server → Both	Force disconnect call

6.2 Agora SDK Integration

Agora RTC SDK is used for all voice and video streaming. The backend generates a short-lived Agora token per session to ensure security.

Token generation (Node.js):

```
const { RtcTokenBuilder, RtcRole } = require("agora-access-token");
const token = RtcTokenBuilder.buildTokenWithUid(
  APP_ID, APP_CERTIFICATE, channelName, uid,
  RtcRole.PUBLISHER, expirationTime
);
```



7. Wallet & Payment System

7.1 Coin Economy

Pack Name	Coins	Price (INR)	Rate
Starter Pack	100 coins	₹99	1 coin = ₹0.99
Basic Pack	300 coins	₹249	1 coin = ₹0.83
Popular Pack	700 coins	₹499	1 coin = ₹0.71
Premium Pack	1500 coins	₹999	1 coin = ₹0.67
VIP Pack	3500 coins	₹1999	1 coin = ₹0.57

7.2 Billing Logic

- On session start: pre-authorise coins = rate × 1 minute
- Every 60 seconds: deduct listener rate from user wallet, credit to listener escrow
- On session end: release partial-minute proration if applicable
- If wallet balance drops below 1-minute cost: send balance:low event
- At zero balance: send balance:zero event and terminate session

7.3 Payment Gateway (Razorpay)

- Create order via Razorpay Orders API on coin pack selection
- Payment captured via Razorpay Checkout (SDK or WebView)
- Webhook: payment.captured → credit coins to user wallet atomically
- Webhook: payment.failed → notify user
- All transactions logged in Payments collection with Razorpay order ID

7.4 Listener Payout

- Listener requests withdrawal from earnings wallet
- Admin reviews and approves withdrawal requests
- Funds transferred via Razorpay Payouts / direct bank transfer
- Platform commission (configurable, e.g. 30%) deducted before payout
- Payout status: Pending → Processing → Completed / Failed



8. Admin Panel

A web-based admin panel (React.js) provides complete governance over the platform.

Module	Capabilities
Dashboard	Real-time KPIs: active sessions, DAU, revenue, new signups
User Management	View, search, ban/unban customer accounts
Listener Management	Approve/reject registrations, set featured status, view earnings
Verification Queue	Review ID documents and selfies; approve or reject with notes
Wallet & Transactions	View all coin purchases, refunds, manual credit/debit
Payout Management	Approve withdrawal requests, view payout history
Commission Settings	Set global or per-listener commission percentage
Content Moderation	Review reported users, review chat logs, issue bans
Reports & Analytics	Session counts, revenue trends, top listeners, retention
Notification Center	Send broadcast push notifications to users/listeners
Configuration	Coin pack pricing, minimum payout, rate limits, app settings





9. Technology Stack

Component	Technology	Notes
Mobile App	Flutter 3.x (Dart) or React Native 0.73	Single codebase for Android & iOS
State Management	Bloc / Provider (Flutter) or Redux (RN)	Predictable state, easy debugging
Backend API	Node.js 20 + Express 4	RESTful API, middleware-based auth
Real-Time	Socket.io 4.x	Bi-directional event-driven comms
Voice/Video	Agora RTC SDK 4.x	Low-latency media streaming
Database	MongoDB 7 (Atlas)	Flexible document schema
Cache	Redis 7 (Upstash)	Presence, rate-limit, token store
File Storage	AWS S3 / Firebase Storage	Profile images, ID documents
Push Notifs	Firebase Cloud Messaging	Cross-platform push delivery
Payments	Razorpay SDK	INR payments & payouts
Auth	JWT + OTP (MSG91)	Stateless auth, mobile OTP
Admin UI	React 18 + Ant Design	Web admin dashboard
Hosting	AWS EC2 / Railway / Render	Scalable cloud deployment
CI/CD	GitHub Actions	Automated build & deploy pipeline
Monitoring	Sentry + Datadog	Error tracking & APM



10. Database Schema

Users Collection

Field	Type	Description
_id	ObjectId	Primary key
role	String	"CUSTOMER" "LISTENER" "ADMIN"
name	String	Display name
mobile	String	E.164 format, indexed unique
email	String	Optional, indexed
profilePhoto	String	S3/Firebase URL
walletBalance	Number	Current coin balance (customers)
earningsBalance	Number	Accumulated earnings in INR (listeners)
isVerified	Boolean	Listener verification status
isOnline	Boolean	Real-time presence
blockedUsers	[ObjectId]	Array of blocked user IDs
createdAt	Date	Account creation timestamp

ListenerProfiles Collection

Field	Type	Description
_id	ObjectId	Primary key
userId	ObjectId	Ref: Users
bio	String	Listener introduction
languages	[String]	Spoken languages
categories	[String]	e.g. ["friendly", "advice"]
chatRate	Number	Coins per minute (chat)
voiceRate	Number	Coins per minute (voice)
videoRate	Number	Coins per minute (video)
avgRating	Number	Computed average (0-5)
totalSessions	Number	Completed session count
photos	[String]	Gallery URLs (max 5)
status	String	"PENDING" "APPROVED" "REJECTED" "SUSPENDED"

Sessions Collection

Field	Type	Description
_id	ObjectId	Primary key
customerId	ObjectId	Ref: Users
listenerId	ObjectId	Ref: Users
type	String	"CHAT" "VOICE" "VIDEO"
status	String	"ACTIVE" "ENDED" "MISSED"
startedAt	Date	Session start time
endedAt	Date	Session end time
durationSeconds	Number	Total call duration
coinsDeducted	Number	Total coins charged to customer
coinsEarned	Number	Coins credited to listener escrow
rating	Number	Post-session rating (1-5)
review	String	Optional text review

Transactions Collection

Field	Type	Description
_id	ObjectId	Primary key
userId	ObjectId	Ref: Users
type	String	"PURCHASE" "DEBIT" "EARNING" "PAYOUT"
amount	Number	Amount in coins or INR
razorpayOrderId	String	For purchases
sessionId	ObjectId	Ref: Sessions (for debits)
status	String	"PENDING" "SUCCESS" "FAILED"
createdAt	Date	Transaction timestamp

11. API Endpoints

Authentication

Method	Endpoint	Description	Access
POST	/api/auth/request-otp	Send OTP to mobile/email	Public
POST	/api/auth/verify-otp	Verify OTP, return JWT	Public
POST	/api/auth/refresh	Refresh access token	Public
POST	/api/auth/logout	Invalidate refresh token	Auth

User & Listener Profiles

Method	Endpoint	Description	Access
GET	/api/users/me	Get current user profile	Auth
PUT	/api/users/me	Update profile	Auth
GET	/api/listeners	List online listeners (paginated)	Auth
GET	/api/listeners/:id	Get listener profile	Auth
POST	/api/listeners/register	Submit listener registration	Auth
PUT	/api/listeners/status	Update online/offline/busy	Listener
PUT	/api/listeners/rates	Update per-minute rates	Listener

Sessions

Method	Endpoint	Description	Access
POST	/api/sessions/initiate	Create session record, get Agora token	Auth
POST	/api/sessions/:id/accept	Listener accepts session	Listener
POST	/api/sessions/:id/reject	Listener rejects session	Listener
POST	/api/sessions/:id/end	End session, trigger billing	Auth
POST	/api/sessions/:id/rate	Submit post-session rating	Customer
GET	/api/sessions	List user's session history	Auth

Wallet & Payments

Method	Endpoint	Description	Access
GET	/api/wallet	Get wallet balance & history	Auth
POST	/api/wallet/create-order	Create Razorpay order for coin pack	Auth
POST	/api/wallet/verify-payment	Verify payment & credit coins	Auth



Method	Endpoint	Description	Access
GET	/api/wallet/transactions	Transaction history	Auth
POST	/api/earnings/withdraw	Listener withdrawal request	Listener
GET	/api/earnings/history	Listener earnings history	Listener

Admin

Method	Endpoint	Description	Access
GET	/api/admin/listeners/pending	List pending verifications	Admin
POST	/api/admin/listeners/:id/approve	Approve listener	Admin
POST	/api/admin/listeners/:id/reject	Reject listener	Admin
POST	/api/admin/users/:id/ban	Ban user/listener	Admin
GET	/api/admin/payouts/pending	List pending withdrawal requests	Admin
POST	/api/admin/payouts/:id/approve	Process payout	Admin
GET	/api/admin/analytics/summary	Platform KPI summary	Admin





12. Security & Compliance

Area	Measure
Transport Security	TLS 1.3 enforced for all API, Socket.io, and media connections
Authentication	JWT HS256 with short expiry; refresh token rotation
Password / OTP	bcrypt for hashed data; OTP valid for 5 minutes, max 3 attempts
Input Validation	Joi schema validation on all API inputs; parameterised queries
Rate Limiting	Redis-backed rate limiter: 60 req/min per IP on public endpoints
File Uploads	File type & size validation; malware scan before storage
ID Verification	Admin-reviewed manual KYC for all listener accounts
18+ Compliance	Age gate on signup; listener DOB verified against ID document
Data Privacy	GDPR-aligned: data minimisation, right to deletion, consent management
Call Privacy	Agora channel tokens expire after session; no raw media stored by default
Abuse Reporting	In-app report flow; admin queue with 24h SLA; auto-suspend on threshold



13. Push Notifications

Event Type	Target	Priority	Description
CALL_INCOMING	Listener	High	Incoming voice/video call request
CHAT_MESSAGE	Customer/Listener	Normal	New chat message received
BALANCE_LOW	Customer	High	Wallet balance below 1-minute cost
SESSION_ENDED	Both	Normal	Session ended with cost summary
VERIFICATION_APPROVED	Listener	Normal	Account approved by admin
VERIFICATION_REJECTED	Listener	Normal	Account rejected with reason
PAYOUT_PROCESSED	Listener	Normal	Withdrawal processed successfully
PROMO_OFFER	Customer	Low	Promotional coin pack offer

14. Deployment Architecture

Component	Platform	Notes
API Server	AWS EC2 (t3.medium) / Railway	Node.js API, Socket.io, min 2 instances
MongoDB	MongoDB Atlas M10+	Managed cluster, automated backups, replication
Redis	Upstash Redis	Serverless Redis, low-latency presence & cache
Media	Agora Cloud (global)	SD-RTN for low-latency call routing
File Storage	AWS S3 + CloudFront CDN	Profile images, secure presigned URLs
Admin Panel	Vercel / Netlify	Static React build, CI/CD auto-deploy
Mobile CI/CD	GitHub Actions + Fastlane	Automated APK/IPA build & store submission
Monitoring	Sentry (errors) + Datadog (APM)	Real-time error & performance tracking

15. Non-Functional Requirements

Requirement	Target / Detail
Performance	API response < 200ms (p95); call setup < 1.5 seconds
Scalability	Horizontal scaling via load balancer; target 10,000 concurrent sessions
Availability	99.9% uptime SLA; multi-AZ deployment



Requirement	Target / Detail
Call Quality	Audio latency < 150ms; video 720p @ 30fps minimum
Data Retention	Chat history: 1 year; Session metadata: 3 years; ID docs: 5 years
Compliance	IT Act 2000 (India), DPDP Act 2023, GDPR (for international users)
Moderation	18+ content policy; AI-assisted flagging + human review queue
Recovery	RTO: 4 hours; RPO: 1 hour; daily automated backups

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User Flow Diagram

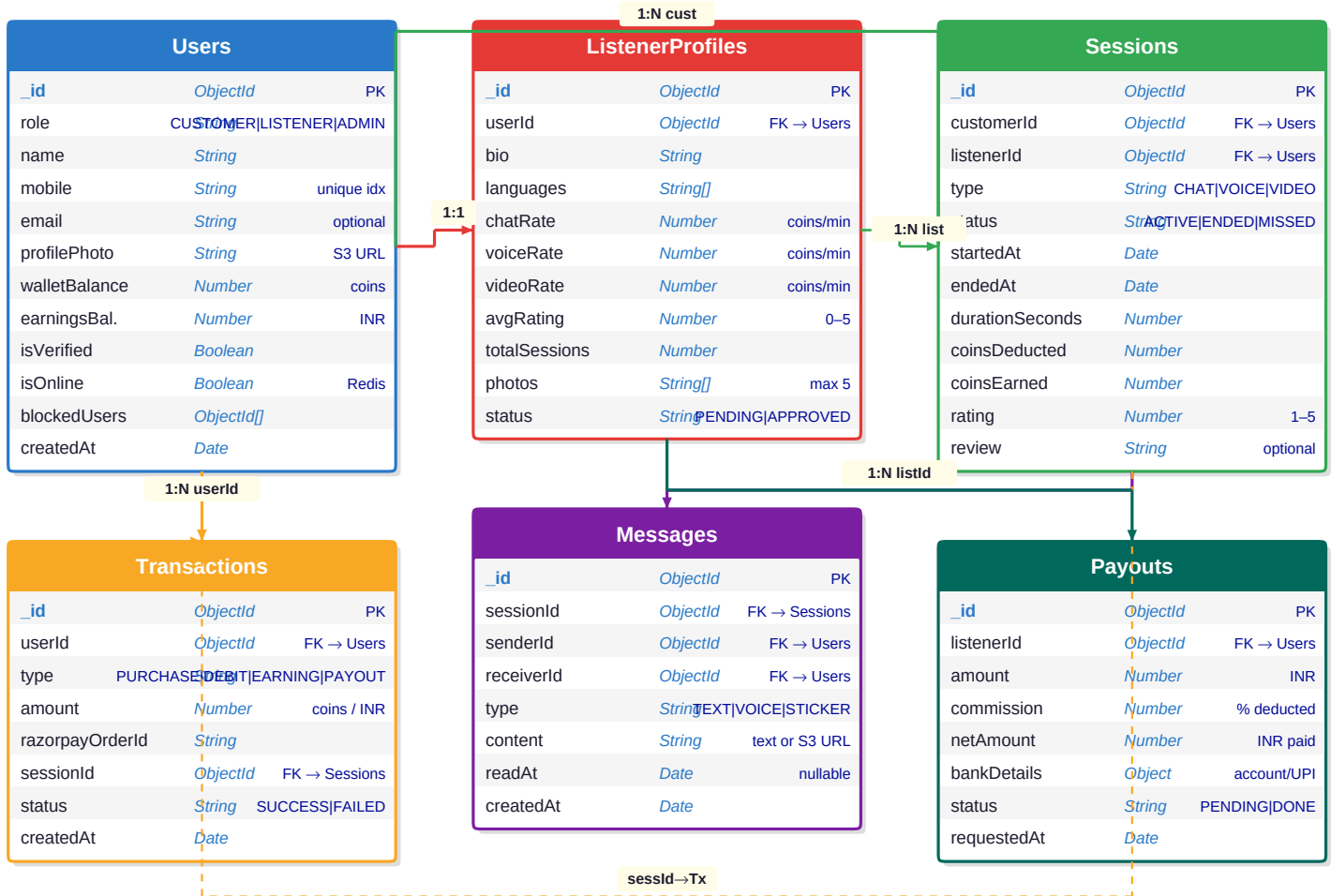
Customer Journey & Listener Journey — End to End





Database ER Diagram

MongoDB Collections & Relationships

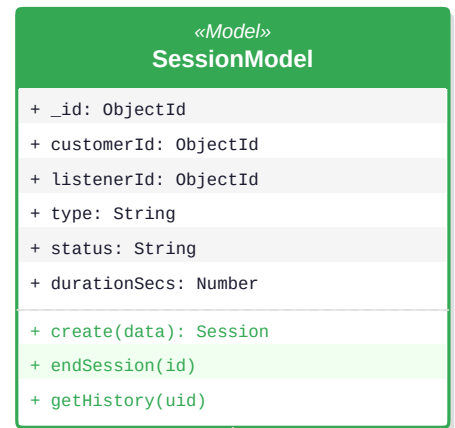
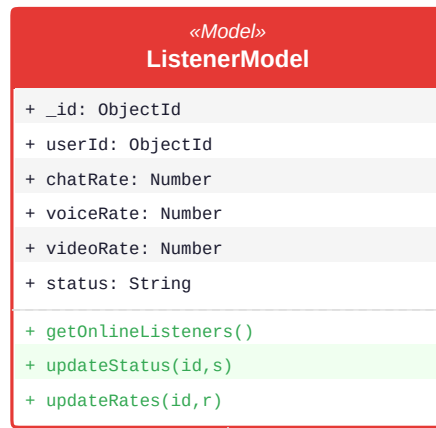
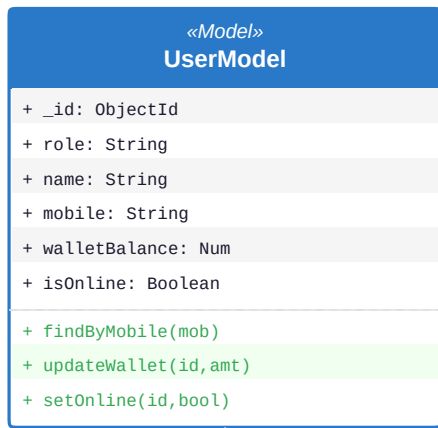




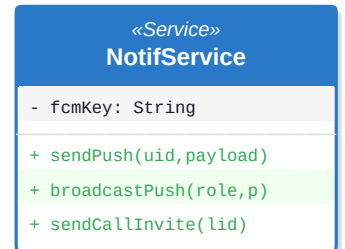
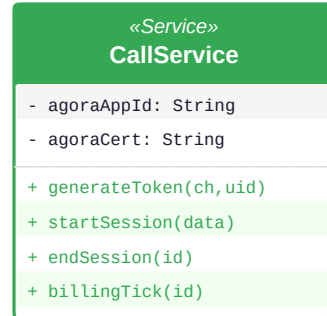
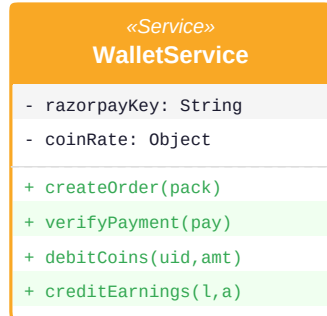
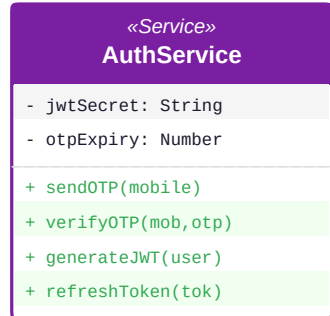
Class Diagram

Backend Models, Services & Controllers (Node.js / MVC Architecture)

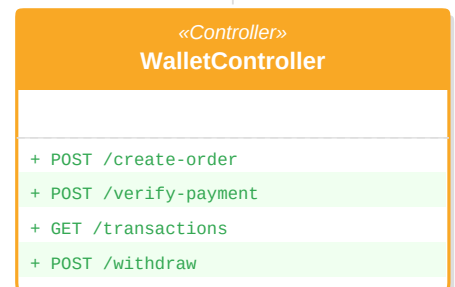
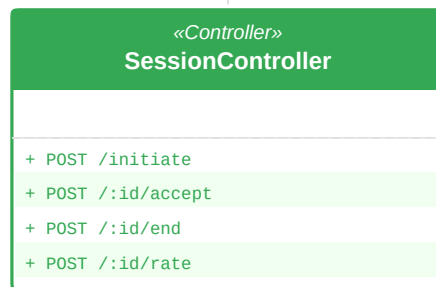
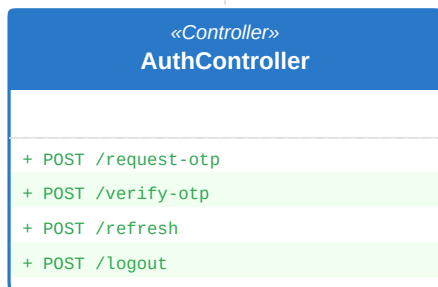
MODELS



SERVICES



CONTROLLERS





Project Layout

Monorepo Folder Structure — Mobile App / Backend / Admin Panel

Mobile App (Flutter)	Backend (Node.js)	Admin Panel (React)
lib/ main.dart app.dart core/ constants/ theme/ utils/ data/ models/ user.dart listener.dart session.dart repositories/ services/ api_service.dart socket_service.dart agora_service.dart presentation/ screens/ auth/ home/ chat/ call/ wallet/ profile/ widgets/ blocs/ l10n/ assets/ images/ fonts/ pubspec.yaml README.md	src/ config/ db.js redis.js agora.js razorpay.js models/ User.js ListenerProfile.js Session.js Transaction.js Message.js Payout.js controllers/ auth.controller.js session.controller.js wallet.controller.js admin.controller.js routes/ auth.routes.js session.routes.js wallet.routes.js admin.routes.js services/ auth.service.js wallet.service.js call.service.js notif.service.js middlewares/ auth.js rateLimit.js validate.js sockets/ chat.socket.js call.socket.js presence.socket.js utils/ app.js server.js .env package.json	src/ api/ adminApi.js authApi.js components/ common/ Header.jsx Sidebar.jsx DataTable.jsx charts/ modals/ pages/ Dashboard.jsx Users.jsx Listeners.jsx Verification.jsx Sessions.jsx Wallet.jsx Payouts.jsx Analytics.jsx Settings.jsx store/ authSlice.js adminSlice.js hooks/ utils/ App.jsx main.jsx index.css vite.config.js tailwind.config.js package.json README.md
■ listener-app/ (monorepo: mobile/ backend/ admin/)		

